Approach B (X/Y), 2D: final implemented method and one-shot confirmation

Verdict: APPROACH_B_2D_CLOSED_PRIMARY_NOT_ESTABLISHED_ORDER_SUPPORTED | EVALUATION_P24.json sha256 d1f6644ceafa9f35

K3 K4: PHYSICALLY VERIFIED and certified necessary in 52 roots (31 K3, 21 K4); learned selection descriptive only

coupling F5: UNSUPPORTED (D - E U -0.010 [-0.031, 0.010])

ordering: SUPPORTED (O2 D - G success on repairable K3/K4 +0.173, Holm CI [0.058, 0.308]); O1/O3 strict-order wins are structural

two d readiness: the 2D pipeline is complete and reproducible; the X/Y advantage over a matched direct predictor is not established

whole system advantage over DIRECT: NOT_ESTABLISHED (U +0.041 / +0.052, Holm-adjusted CIs include 0; below the +0.10 gain gate; unnecessary-repair guardrail would fail: +0.182)

Scope: new seeds of the same 2D K34 doorway generator family; one locked look at 237 independent roots with frozen 15-member ensembles; no cross-generator or 3D claim. Exploratory P25 ablations are recomputations on the labelled cohort, not tests.

0132 final findings -- 1. Final answers

Question	Answer from the independent P24 cohort
Does the X/Y system beat a matched direct success predictor?	Not established. U: D 0.259, E 0.269, DIRECT-GEO 0.218. D − DIRECT +0.041 (Holm CI [−0.005, 0.088]); E − DIRECT +0.052 (Holm CI [0.000, 0.104], p 0.0170 vs a Holm level of 0.0167). Both are below the +0.10 gain gate, and both would fail the unnecessary-repair guardrail (+0.182).
Does order-aware X matter?	Yes. O2: D beats its order-invariant twin G by +0.173 in success on repairable K3/K4 roots (Holm CI [0.058, 0.308]). On 41 certified order-twin roots, D and E rank every twin correctly. G's 0.5 there is structural.
Does the quadratic coupling F5 help?	No. D − E is −0.010 [−0.031, 0.010] on U, and has never been positive in any study.
K3 / K4?	Physically established: 31 K3 and 21 K4 roots are witnessed *and* certified necessary in independent data (K4 was 7 in development). Learned selection on them is descriptive only: K4 success D 0.71, E 0.71, DIRECT 0.52, G 0.62, n = 21.
Is the 2D method ready?	The 2D pipeline is complete, reproducible and custody-verified. The predicted *advantage* of X/Y over a strong direct baseline is small, consistent in sign, and unconfirmed. If X/Y is used, prefer E.

0132 final findings -- 3. Confirmatory results

Arm	U (193 INF)	C (75 AVAIL)	REP34 (52)	Unnecessary (44 FEA)	K0 no-op correct
D XYQ-GEO-SYM	0.259 [0.202, 0.325]	0.667	0.750	0.227	0.773
E XY-UNC-SA-GEO-SYM	0.269 [0.212, 0.336]	0.693	0.769	0.227	0.773
DIRECT-GEO	0.218 [0.165, 0.281]	0.560	0.635	0.045	0.955
G order-invariant	0.212 [0.161, 0.275]	0.547	0.577	0.205	0.795
Analytic access (not learned)	0.212	0.547	—	0.000	1.000
Verify-first oracle (evaluator-only)	0.389	—	—	—	—

0132 final findings -- 3. Confirmatory results

Study	D	E	DIRECT	D − DIRECT
P21 development (398 roots, out-of-fold, 3 seeds)	0.289	0.287	0.258	+0.031 [−0.002, 0.064]
P24 independent (237 roots, frozen 15-member ensembles)	0.259	0.269	0.218	+0.041 [0.000, 0.083] (Holm [−0.005, 0.088])

0132 final findings -- 4. Why the primary failed (failure explanations)

Arm	Success	No-op	X fails during a relocation	X fails at the connector	Y fails	Unverified
D	50	24	86	1	32	0
E	52	24	81	1	34	1
DIRECT	42	20	97	1	32	1
G	41	26	98	1	26	1

0132 final findings -- 5. Claims matrix

#	Claim	Status	Evidence
1	The X/Y factorization (D or E) beats a matched direct success predictor on U	NOT SUPPORTED (not established)	P24 primary; P21 development same direction, also not established
2	X/Y beats it on availability-conditioned C	Consistency only: E − DIRECT +0.133 [0.040, 0.240] raw not a registered claim	P24 C check
3	Order-aware F2 improves outcomes over an order-invariant control on repairable K3/K4 roots	SUPPORTED (confirmatory)	P24 O2 +0.173, Holm CI [0.058, 0.308]
4	F2 orders certified order twins correctly	SUPPORTED (descriptive; G's 0.5 is structural)	41 certified roots: 1.000
5	Quadratic coupling F5 (D vs E) helps	UNSUPPORTED	P24 D − E −0.010; P21 +0.002
6	The X/Y arms keep false repairs at DIRECT's level	NOT SUPPORTED	Unnecessary repairs 10/44 vs 2/44 (+0.182 [0.068, 0.295]), all neutral
7	Learned arms avoid harmful edits on feasible roots	Observed 0 / 44 for all arms. Not proven safe (small denominator)	P24 FEA
8	K3 / K4 necessity exists physically in this 2D family	SUPPORTED	31 K3 and 21 K4 roots witnessed and certified (K_cert = verified K)
9	Learned selection works on K4 specifically	Descriptive only: D 0.71 [0.50, 0.86] vs DIRECT 0.52 [0.32, 0.72], n = 21	No registered per-stratum test
10	The learned scorer adds value over analytic geometry	Weakly: DIRECT (0.218) and G (0.212) are about equal t analytic access (0.212); D/E are about +0.05 above it	P24 arms (no registered test)
11	Y0 (target feasibility) is predicted well	Yes, descriptively: AUROC 0.960–0.967 for all arms	P24 calibration
12	The ensemble is reproducible and in custody	SUPPORTED	Drive read-back 126/126 hashes; P25 re-prediction max difference 0.0 vs the sealed means
13	Results generalize to other generators or to 3D	NOT CLAIMED	Same 2D K34 doorway family, new seeds only
14	The model generates trajectories or discovers global causes	NOT CLAIMED	It ranks a supplied catalogue; the trajectories are the teacher's

0132 final findings -- 6. Exploratory ablations (P25; no claims)

Arm	Range	Median
D	0.223–0.285	0.249
E	0.228–0.280	0.259
DIRECT	0.083–0.249	0.223
G	0.130–0.233	0.202

0132 final findings -- 9. Gate record

Gate	Commits
P23.1	be8b272f → e2c90cde
P24	a7efb1d4 (authorization), 3457e5f4 (seal), 0eb0f390 (labels), 4ba3f0a1 (evaluation), 41636139 (report), ab0f86ed (log paths), 07633a33 (witness files)
P25	the commit that adds this file; the SHA is reported in the hand-off message

final implemented methodology -- 3. Scorer (original F1–F5, with the P17 extensions GEO2 and SYM)

Factor	Role
F1	analytic admissibility screen (never learned)
F2	X unary a_r for the ordered prefix; predecessor-aware, so order matters
F3	Y unary b_s for the edited scene
F4	omissions, scored through the same F2 / F3
F5	optional coupling w_r; the uncoupled variant E fixes w = 0

final implemented methodology -- 3. Scorer (original F1–F5, with the P17 extensions GEO2 and SYM)

Arm	No-op gate	α	γ
D	0.7	0	0.05
E	0.7	0	0.05
DIRECT-GEO	0.5	0.3	0
G	none	0	0.05

final implemented methodology -- 3. Scorer (original F1–F5, with the P17 extensions GEO2 and SYM)

Arm	Description
D XYQ-GEO-SYM	full X/Y with quadratic coupling
E XY-UNC-SA-GEO-SYM	X/Y, w = 0
DIRECT-GEO	one success head on the same inputs, no X/Y split
G XYQ-GEO-SYM-ORDINV	D with an order-invariant F2

final implemented methodology -- 4. Difficulty and necessity

Term	Meaning	Evidence status
K_intent	the generator's request	never evidence
K_cert	every set of K − 1 movable discs, when removed, leaves the target's declared pick or place modes certified closed	sound necessity bound
Verified K	the fewest moves of a teacher-verified candidate, with every shorter catalogue candidate decided	—
Certified-necessary K	K_cert = verified K	—

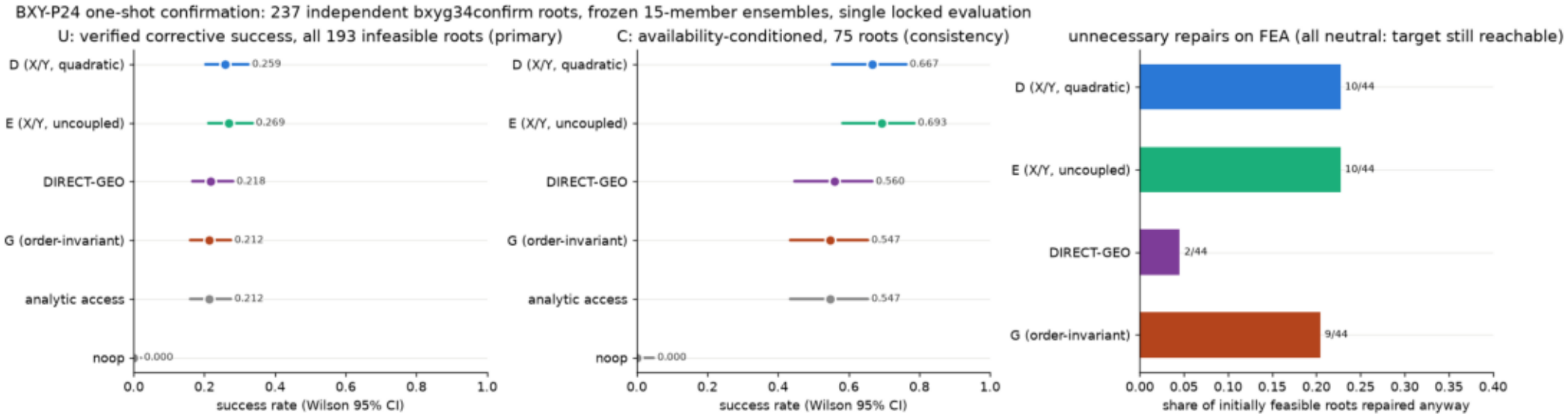
final implemented methodology -- 5. Confirmatory protocol (P23 lock → P24 one-shot)

Family, K_intent	Slots
local_mixed, K2	22
local_mixed, K3	66
local_mixed, K4	198
local_pick, K3	44
local_pick, K4	44
v21b_near_closed, K0	66

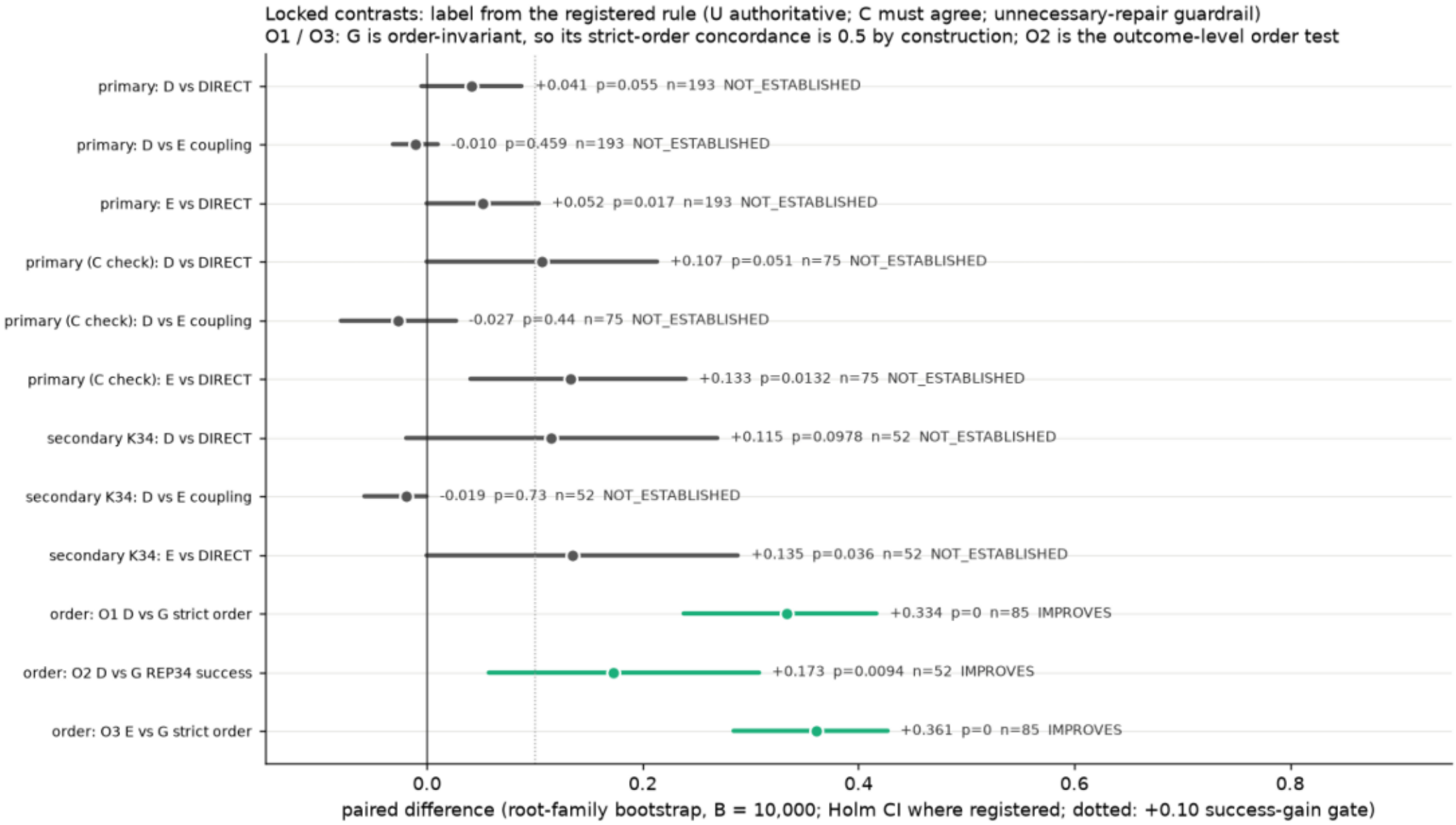
final implemented methodology -- 6. Result (P24, one look)

Arm	U
D	0.259
E	0.269
DIRECT-GEO	0.218
G	0.212

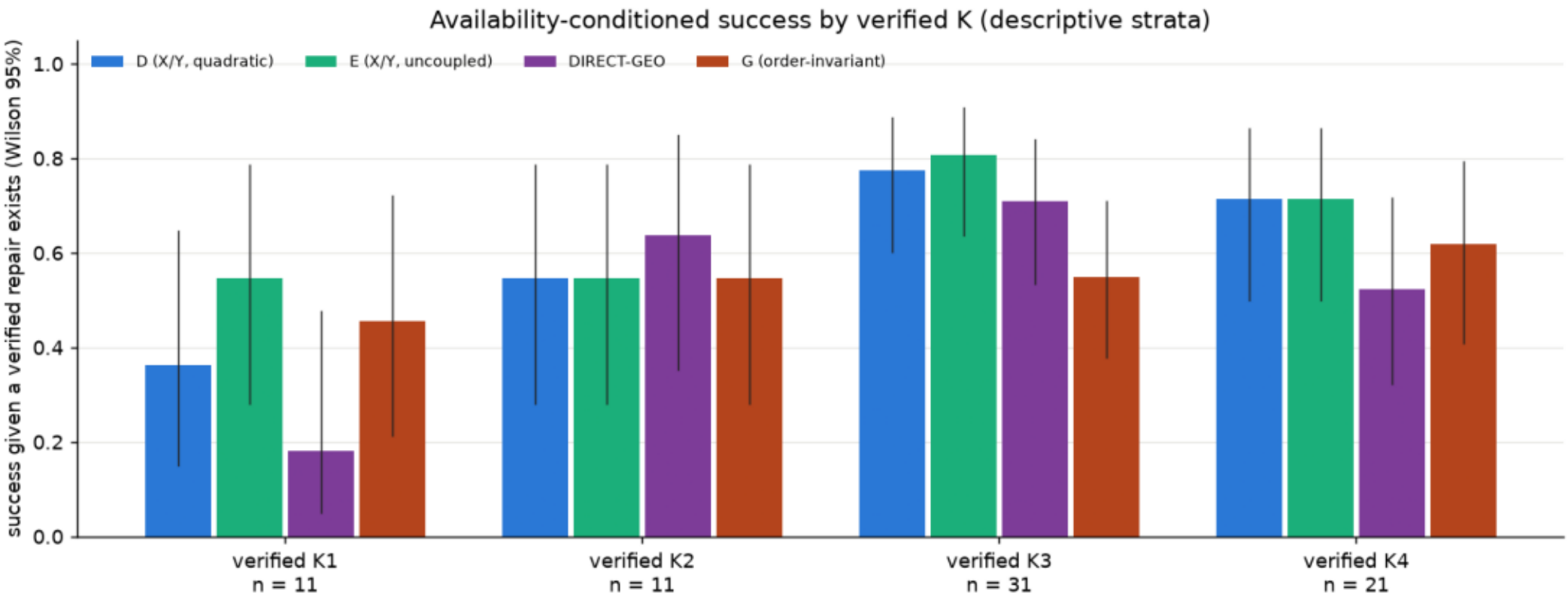
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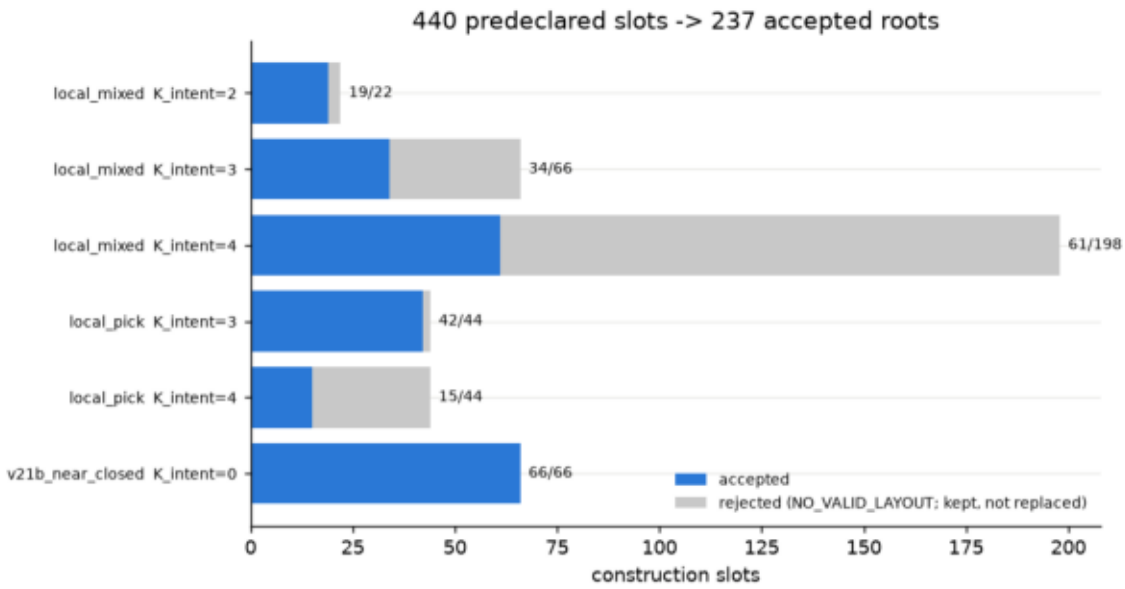
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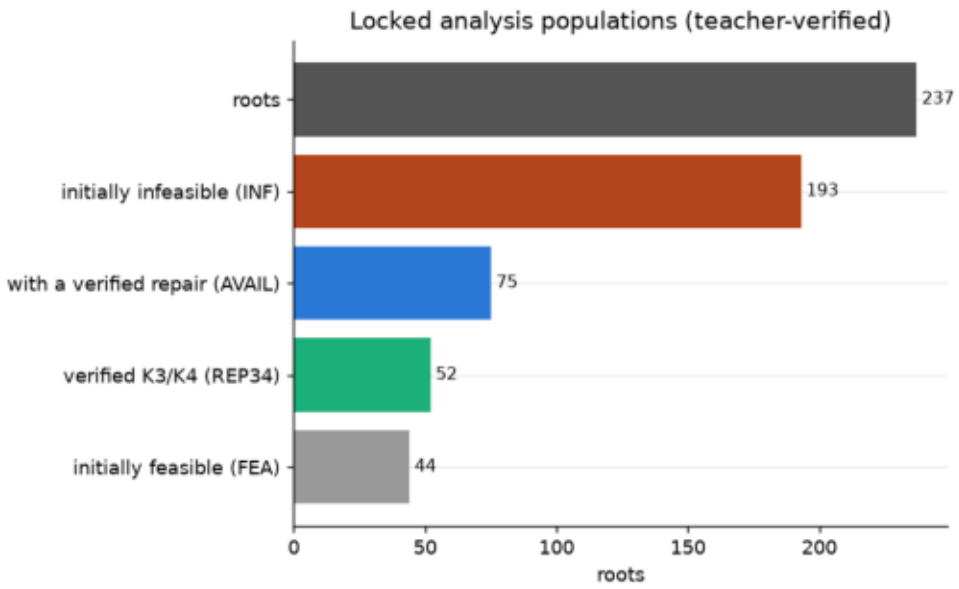


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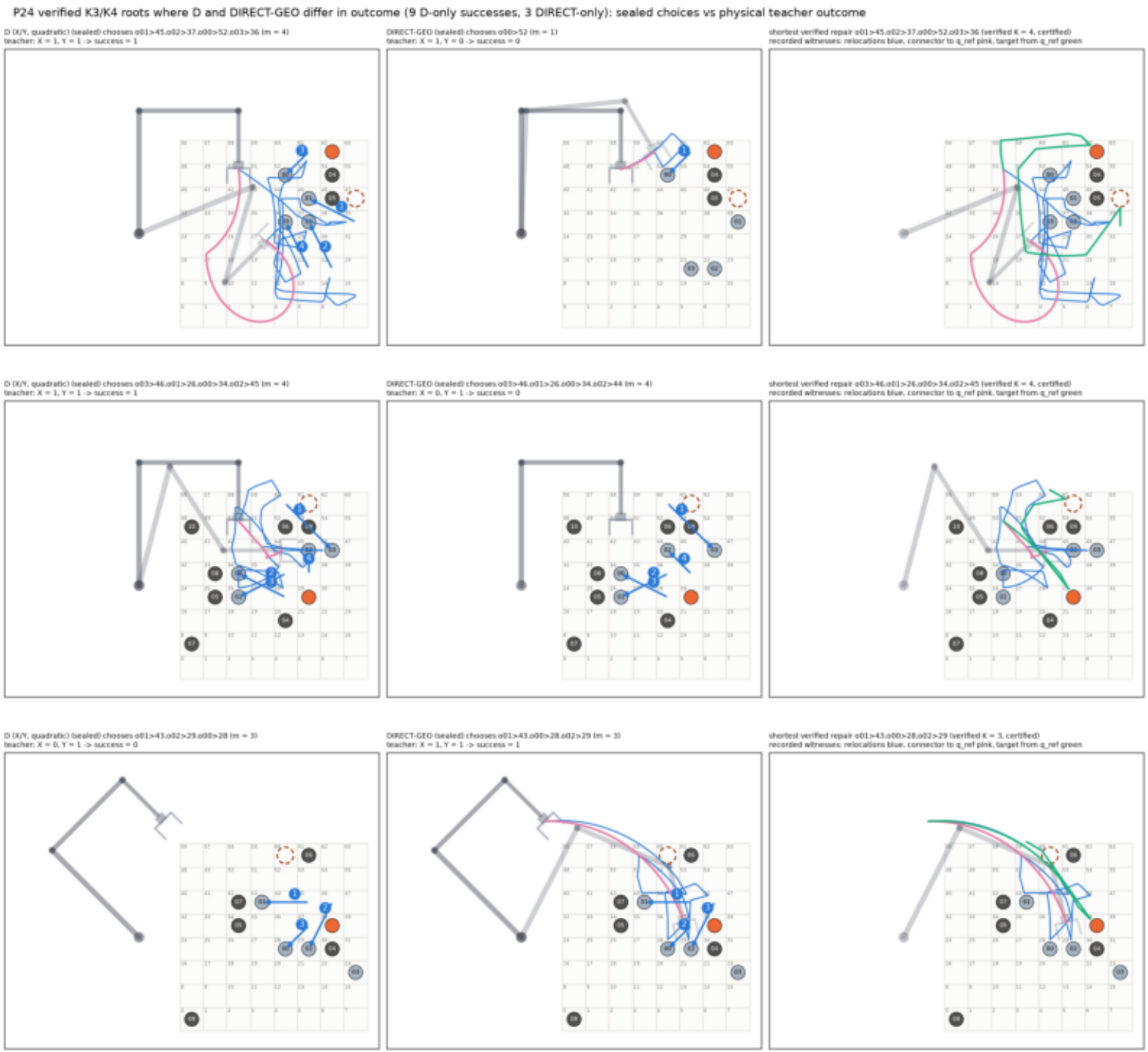


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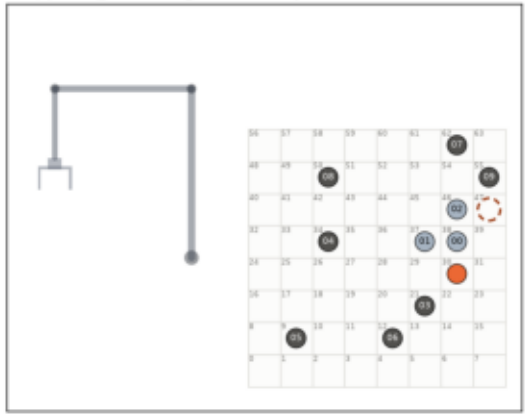
visualizations/approach_b/bxy_p24_v1/05_choices_D_vs_DIRECT.png



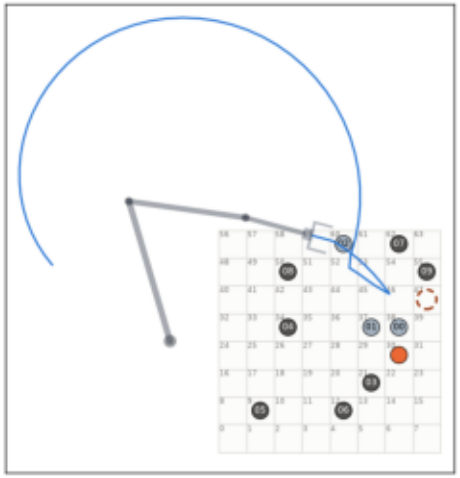
visualizations/approach_b/bxy_p24_v1/06_storyboard_K3_s025.png

2d-bxyg34confirm-c34-s025: verified repair o02>60.o01>28.o00>39 (verified K = 3), every panel drawn from recorded teacher witnesses

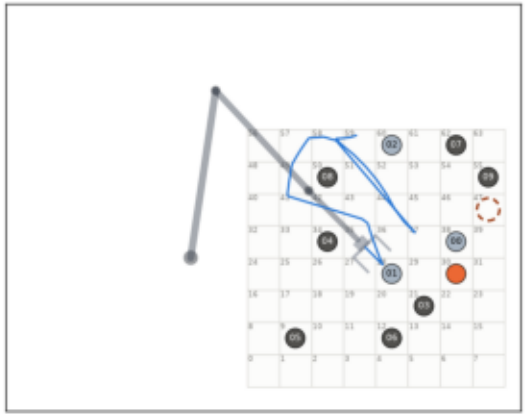
0. initial scene, robot at q0
Yempty) from q_ref: CERTIFIED_INFEASIBLE -> target blocked



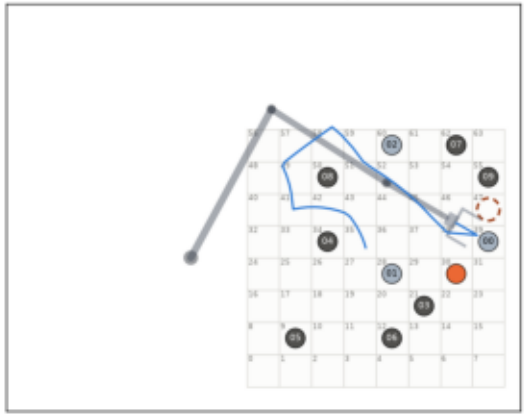
1. relocate o02 -> cell 60
recorded witness of this relocation (online); starts where step 0 ended (no reset)



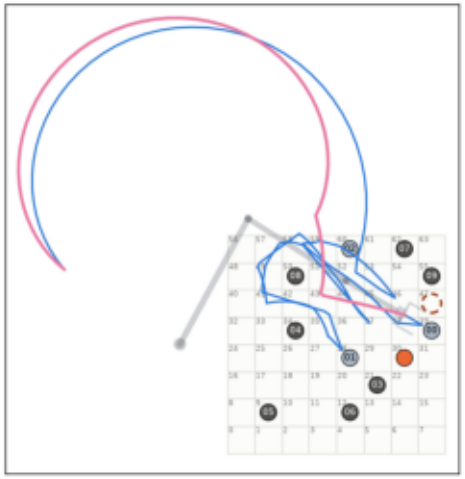
2. relocate o01 -> cell 28
recorded witness of this relocation (online); starts where step 1 ended (no reset)



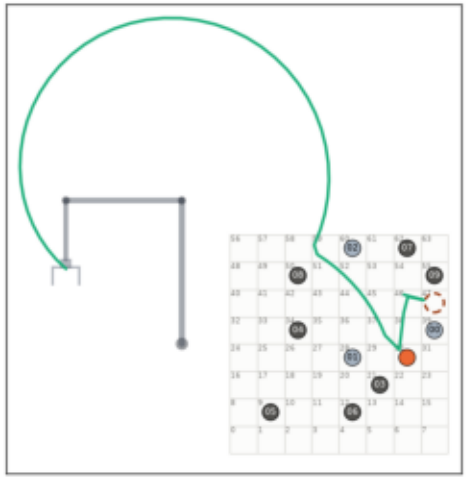
3. relocate o00 -> cell 39
recorded witness of this relocation (online); starts where step 2 ended (no reset)



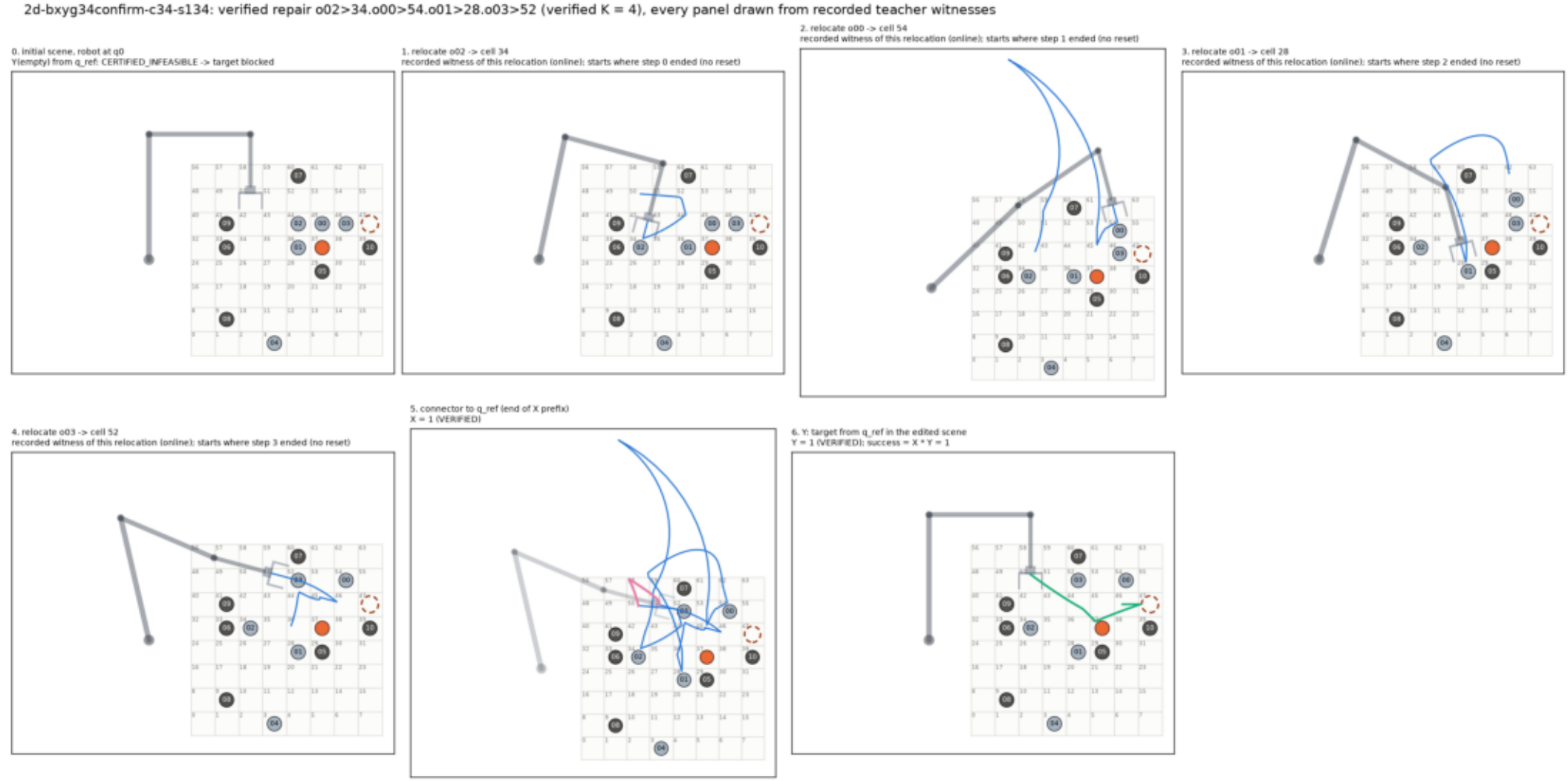
4. connector to q_ref (end of X prefix)
X = 1 (VERIFIED)



5. Y: target from q_ref in the edited scene
Y = 1 (VERIFIED); success = X * Y = 1

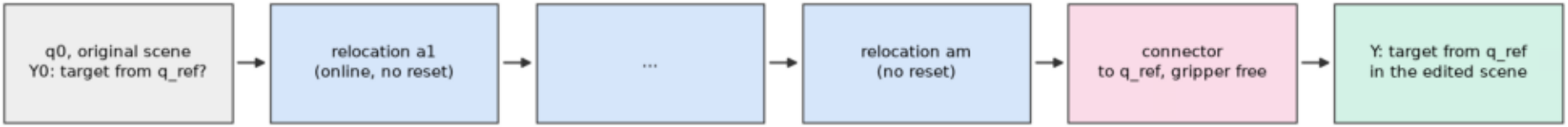


visualizations/approach_b/bxy_p24_v1/06_storyboard_K4_s134.png



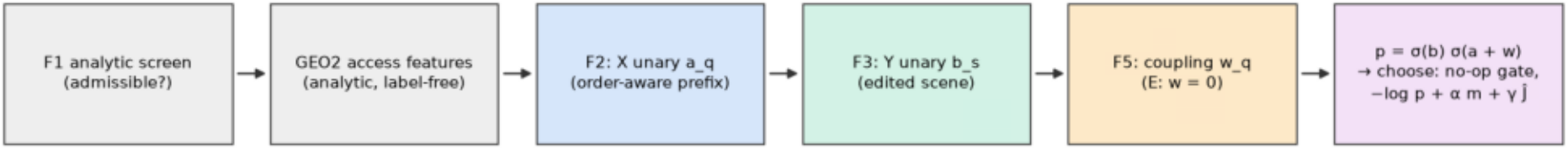
visualizations/approach_b/bxy_p25_v1/00_protocol_and_architecture_schematic.png

Teacher protocol (labels)



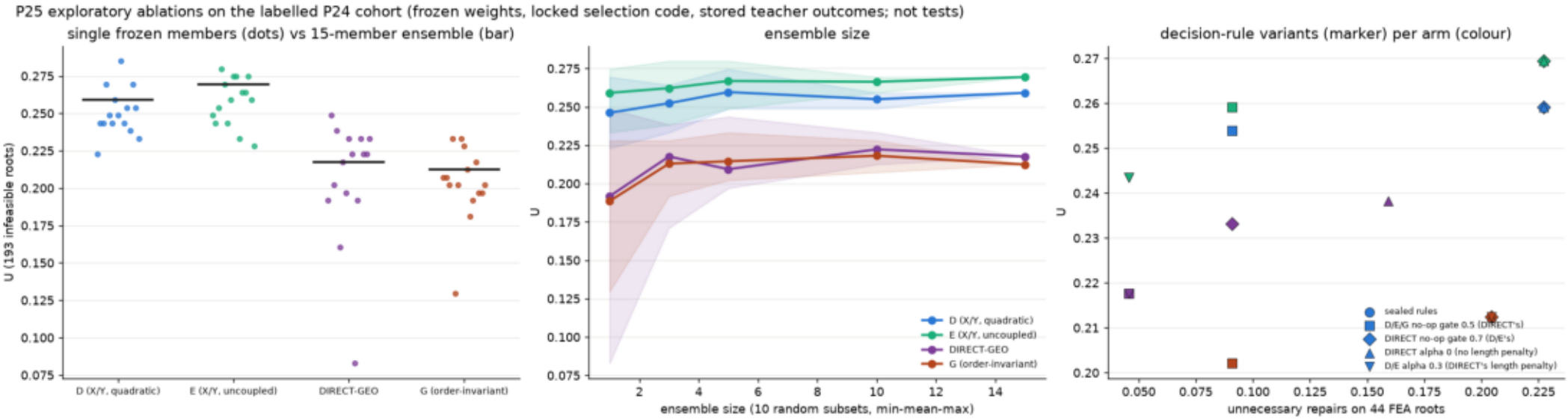
X_q = 1 iff every relocation and the connector execute (prefix order matters); Y_s = 1 iff the target succeeds from q_ref in the edited scene; success = X * Y; negatives keep CERTIFIED vs NO_PLAN_FOUND/TIMEOUT evaluator-side

Scorer (frozen; ranks an already-generated catalogue; never generates trajectories)



DIRECT-GEO: one success head on the same inputs (no X/Y split). G: F2 sees no order (order-invariant control). Training: joint X/Y likelihood + travel (Huber) + SYM edit contrast; 15 frozen members per arm (P21).

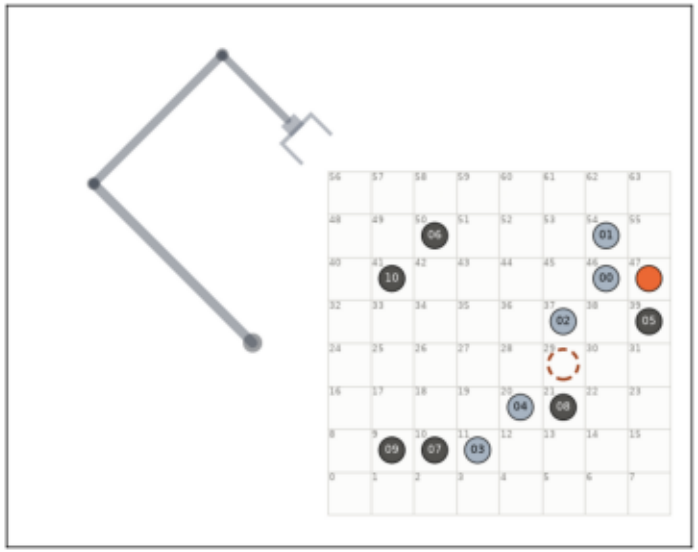
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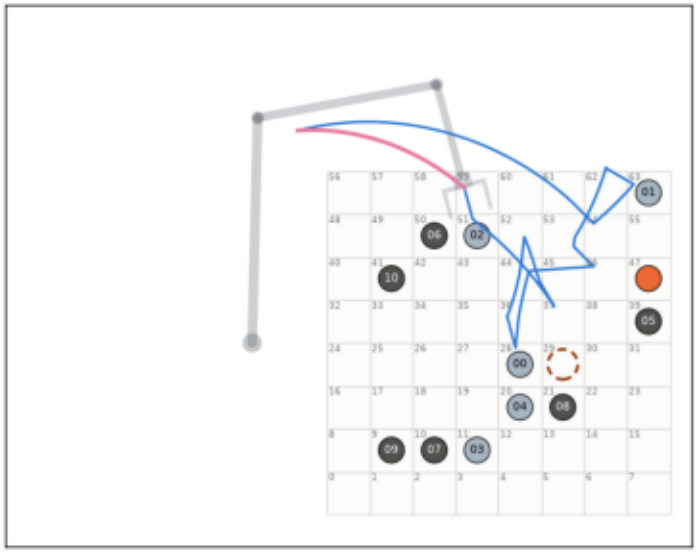
visualizations/approach_b/bxy_p25_v1/02_certified_order_twin.png

Certified order twin at 2d-bxyg34confirm-c34-s062: same moves and endpoint, different order -> different X (sealed successes D 1, G 0)

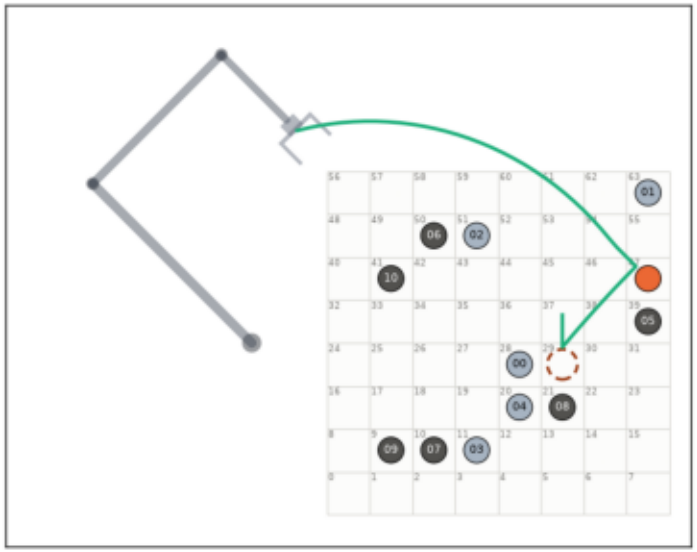
order o00>28.o01>63.o02>51
X = 0 at relocation:0: CERTIFIED_INFEASIBLE (certified); 0 relocation(s) done



order o01>63.o00>28.o02>51
X = 1: relocations blue, connector pink



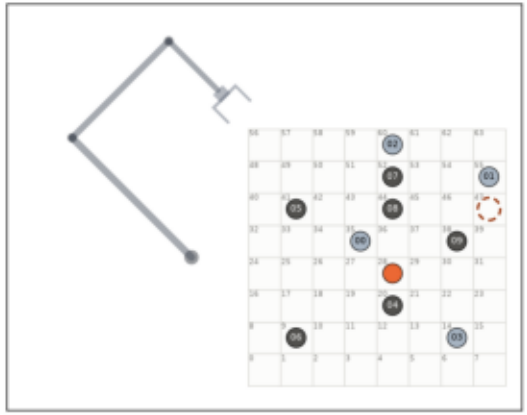
shared Y: same edited scene, target from q_ref
Y = 1 (VERIFIED)



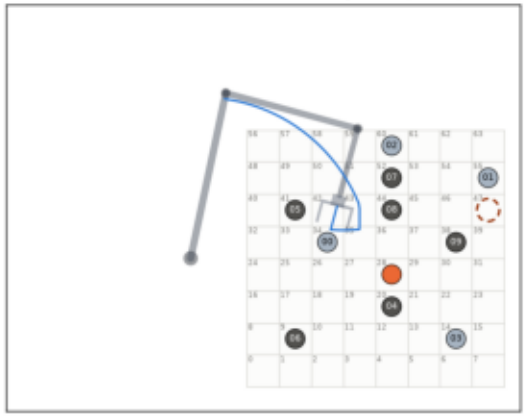
visualizations/approach_b/bxy_p25_v1/03_storyboard_K3_s026_D_choice.png

2d-bxyg34confirm-c34-s026: verified repair o00>34.o02>53.o01>63 (verified K = 3), every panel drawn from recorded teacher witnesses

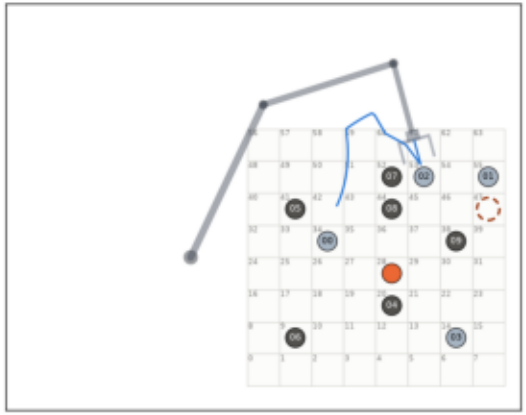
0. initial scene, robot at q0
Y(empty) from q_ref: CERTIFIED_INFEASIBLE -> target blocked



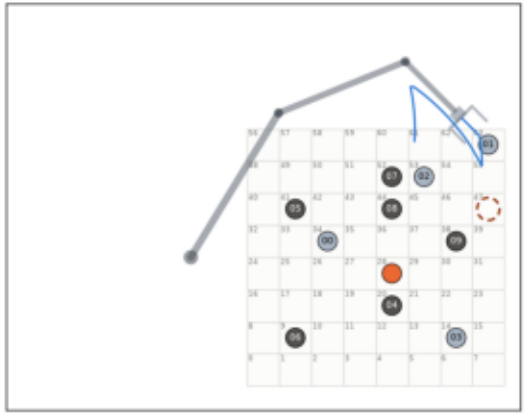
1. relocate o00 -> cell 34
recorded witness of this relocation (online): starts where step 0 ended (no reset)



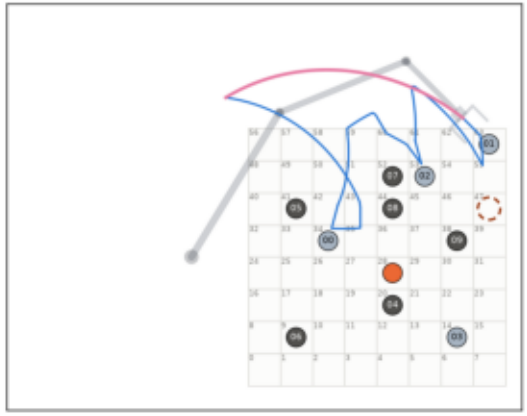
2. relocate o02 -> cell 53
recorded witness of this relocation (online): starts where step 1 ended (no reset)



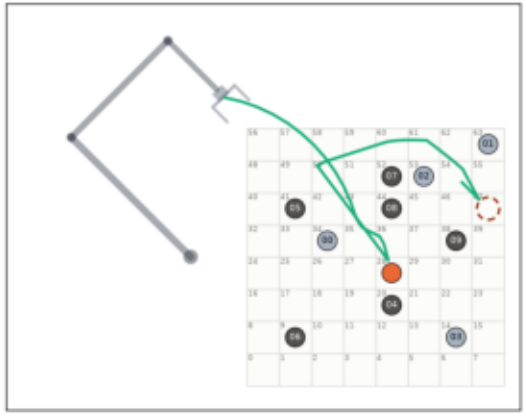
3. relocate o01 -> cell 63
recorded witness of this relocation (online): starts where step 2 ended (no reset)



4. connector to q_ref (end of X prefix)
X = 1 (VERIFIED)



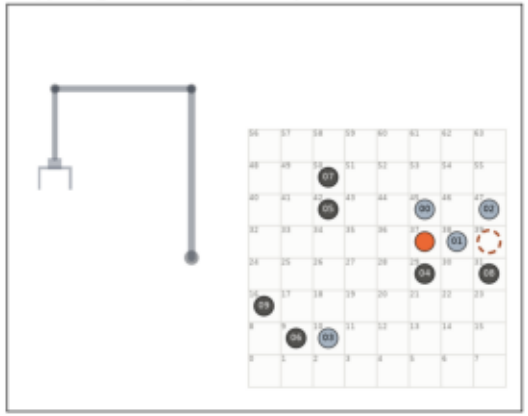
5. Y: target from q_ref in the edited scene
Y = 1 (VERIFIED): success = X * Y = 1



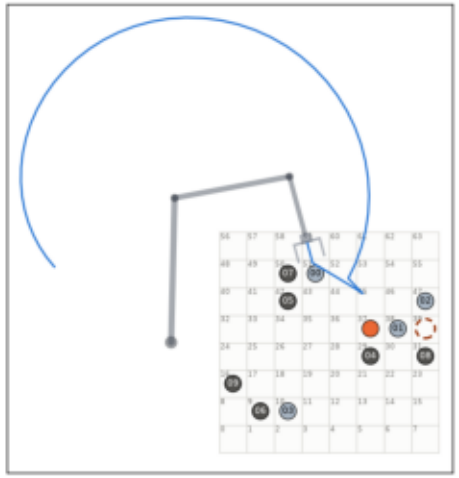
visualizations/approach_b/bxy_p25_v1/03_storyboard_K3_s029_D_choice.png

2d-bxyg34confirm-c34-s029: verified repair o00>51.o01>28.o02>55 (verified K = 3), every panel drawn from recorded teacher witnesses

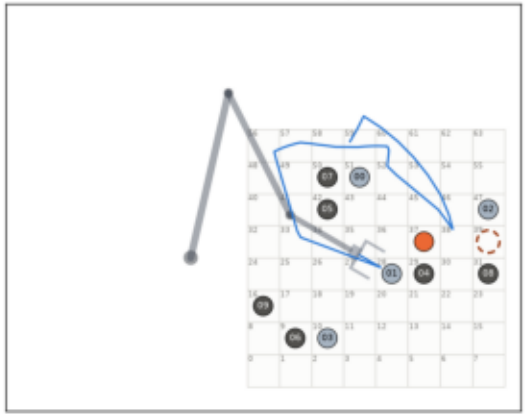
0. initial scene, robot at q0
Yl(empty) from q_ref: CERTIFIED_INFEASIBLE -> target blocked



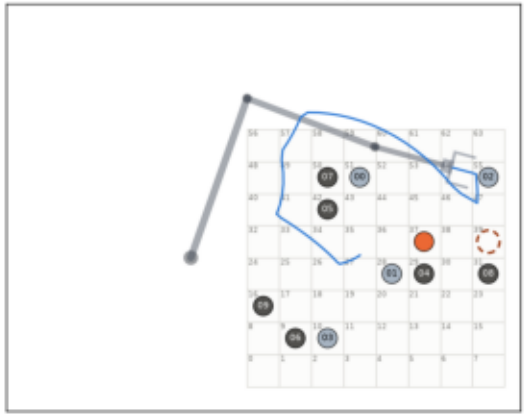
1. relocate o00 -> cell 51
recorded witness of this relocation (online); starts where step 0 ended (no reset)



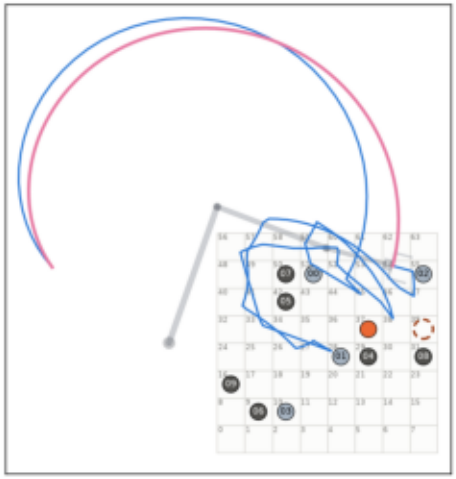
2. relocate o01 -> cell 28
recorded witness of this relocation (online); starts where step 1 ended (no reset)



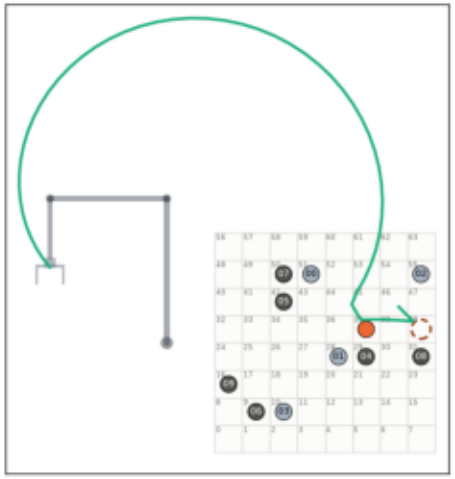
3. relocate o02 -> cell 55
recorded witness of this relocation (online); starts where step 2 ended (no reset)



4. connector to q_ref (end of X prefix)
X = 1 (VERIFIED)



5. Yl target from q_ref in the edited scene
Y = 1 (VERIFIED); success = X * Y = 1

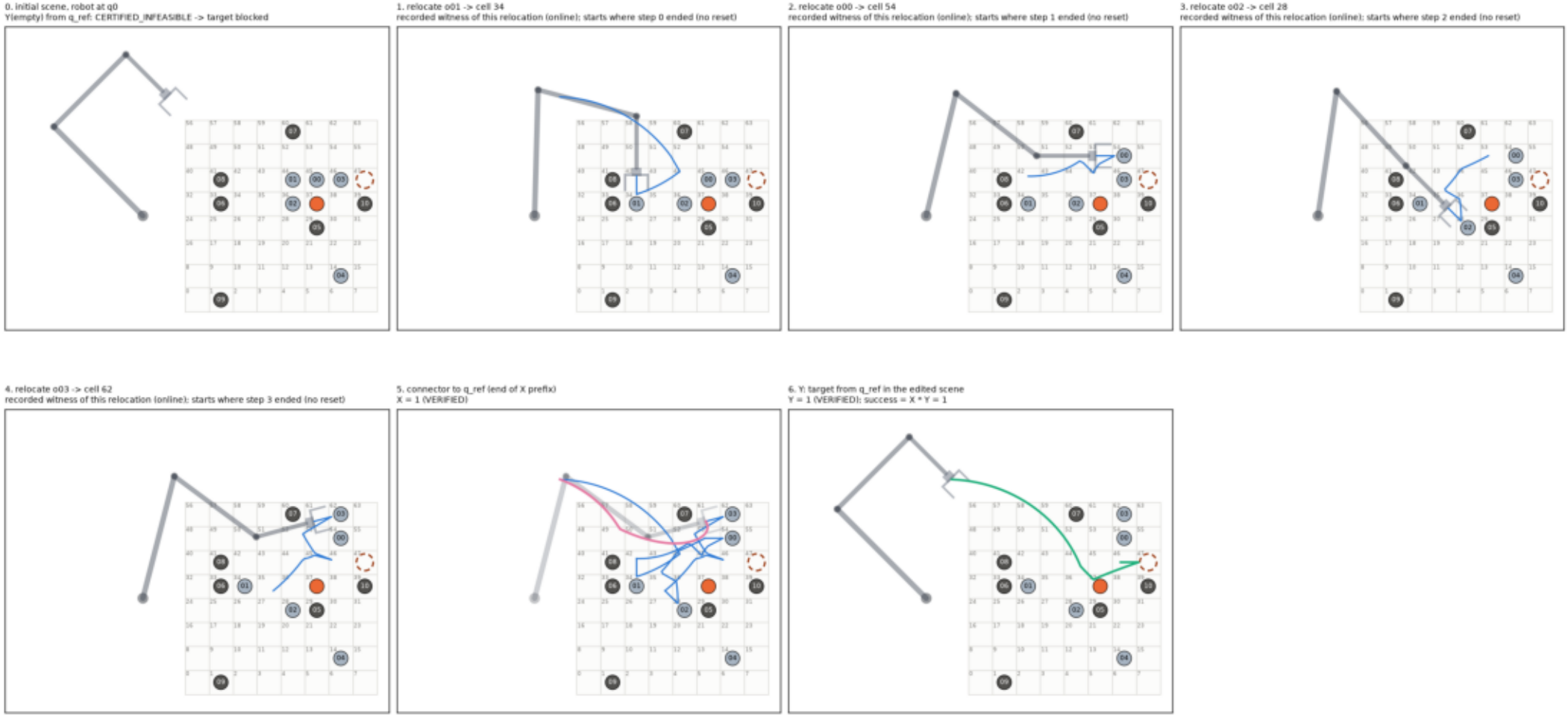


visualizations/approach_b/bxy_p25_v1/03_storyboard_K4_s134_D_choice.png

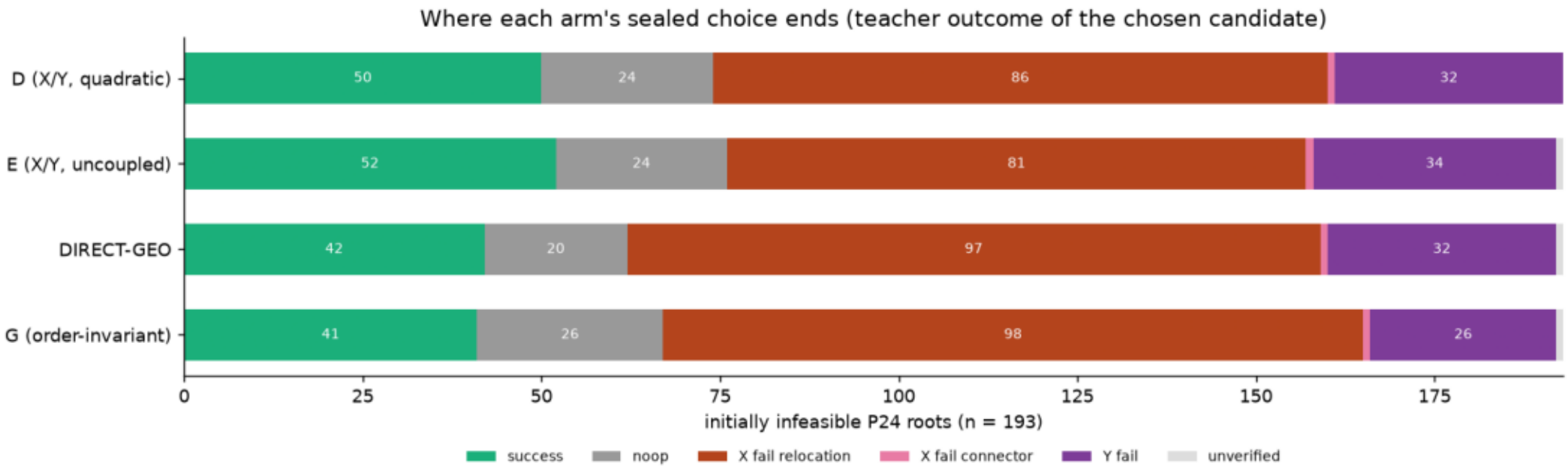


visualizations/approach_b/bxy_p25_v1/03_storyboard_K4_s143_D_choice.png

2d-bxyg34confirm-c34-s143: verified repair o01>34.o00>54.o02>28.o03>62 (verified K = 4), every panel drawn from recorded teacher witnesses



visualizations/approach_b/bxy_p25_v1/04_failure_breakdown.png



visualizations/approach_b/bxy_p25_v1/05_frontier.png

